# Pablo Aparicio Martín - Portfolio

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## Project 1: Integrated Biomedical Platform (BSc Industrial Design Thesis)

Developed for the 'TALES' biomedical research project, this platform required the precise mechanical integration and packaging of multiple sensors (optical, electronic, and chemical) around a highly delicate microfluidic cell-on-a-chip.

I managed the end-to-end product development cycle: from initial conceptualization and parametric 3D CAD modeling, to material selection and hands-on fabrication using my own FFF/FDM 3D printing setup. Additionally, I created comprehensive technical drawings optimized for CNC machining to ensure scalability for future manufacturing updates. Those are available in this link.

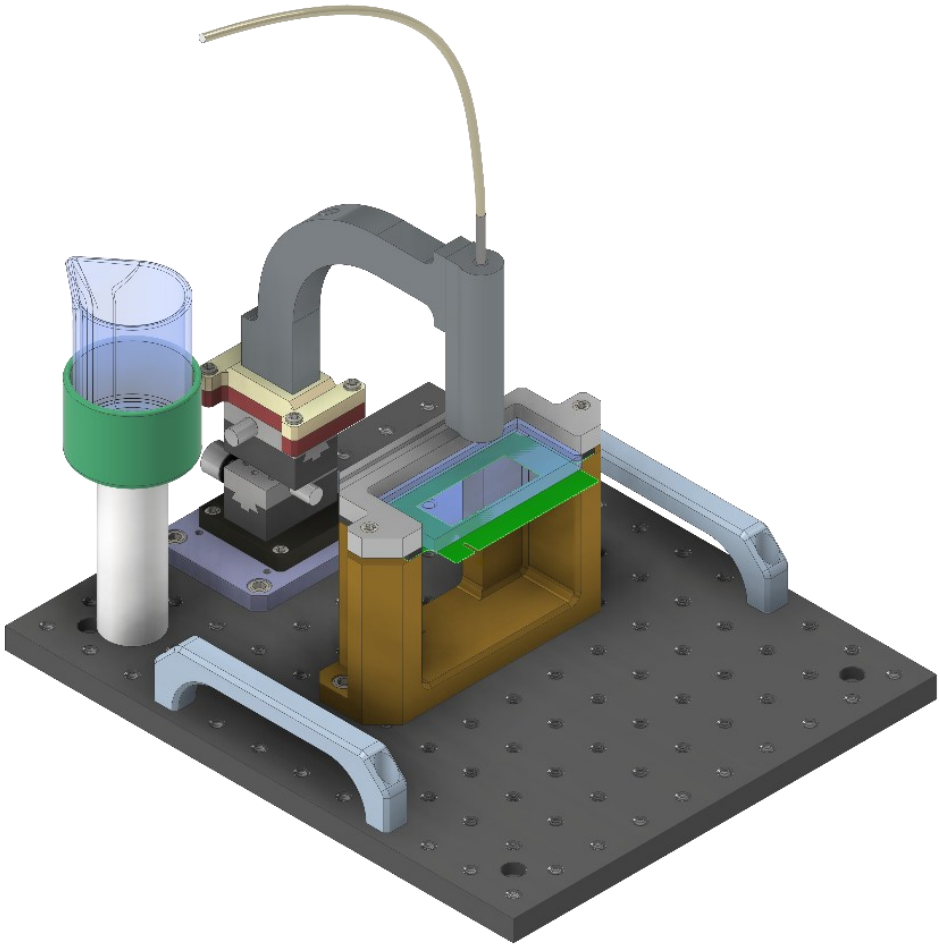


*Fig. 1. 3D CAD assembly.*



*Fig. 2. Functional platform components fabricated via FDM in PETG.*

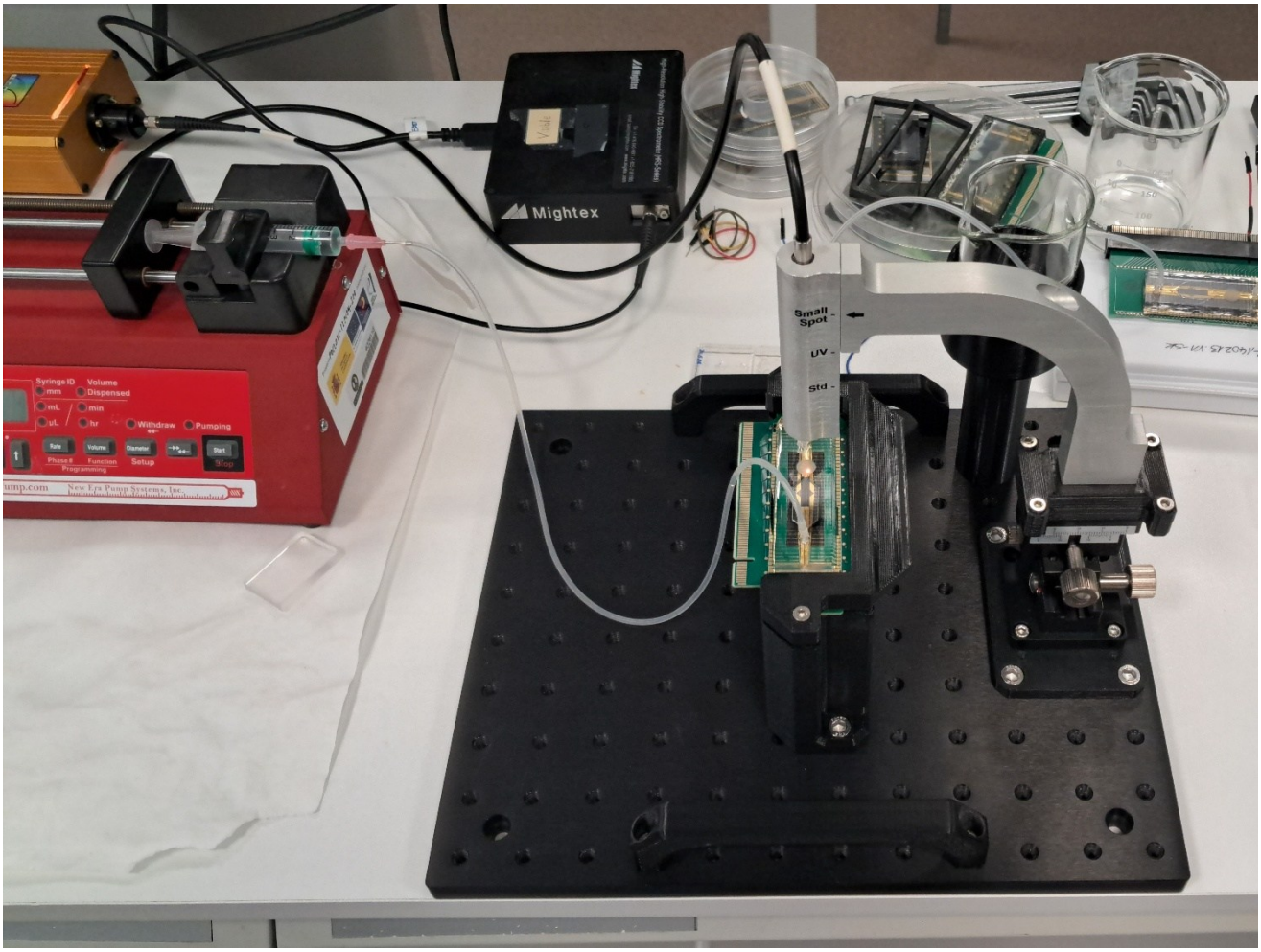



*Fig. 3. Fully assembled biomedical platform actively conducting optical measurements in the laboratory.*

## Project 2: Sustainable Solar Cooker (Workshop Fabrication & Rapid Prototyping)

Working within an engineering team to develop a functional solar cooker from repurposed materials, I was specifically responsible for core mechanical design features, creating the technical manufacturing drawings, and executing the hands-on workshop fabrication. Moving beyond digital prototyping, this project required manual mechanical assembly (precise drilling, cutting, structural joining, and material adaptation) demonstrating my pragmatic problem-solving attitude and proficiency with standard power tools to rapidly turn concepts into testable hardware.

My team's solar cooker resulted as the best one in the whole class, being able to heat up 0.5L of water up to 70 °C in just 11.5 minutes, while the original requirement was reaching 60 °C in 15 minutes.

To have a further view of the design, in these links you can check out the technical drawings of the prototype and a video of me and a teammate assembling the whole kitchen:

- Drawings
- Assembly



*Fig. 4. Solar cooker fully assembled in the workshop.*



*Fig. 5. Solar cooker installed at the rooftop of the university, heating up the water.*

## Project 3: Custom Automotive Dashboard Integration (Personal Project)

To upgrade my vehicle with a modern Android Auto smart display without losing the legacy onboard computer (which displays critical fuel data and major system alerts), I reverse-engineered the dashboard's geometry to design a custom dual-screen adapter.

This project required precise CAD modeling to ensure a seamless mechanical fit into the existing radio slot, securely packaging both the new smart display and the original OEM screen.

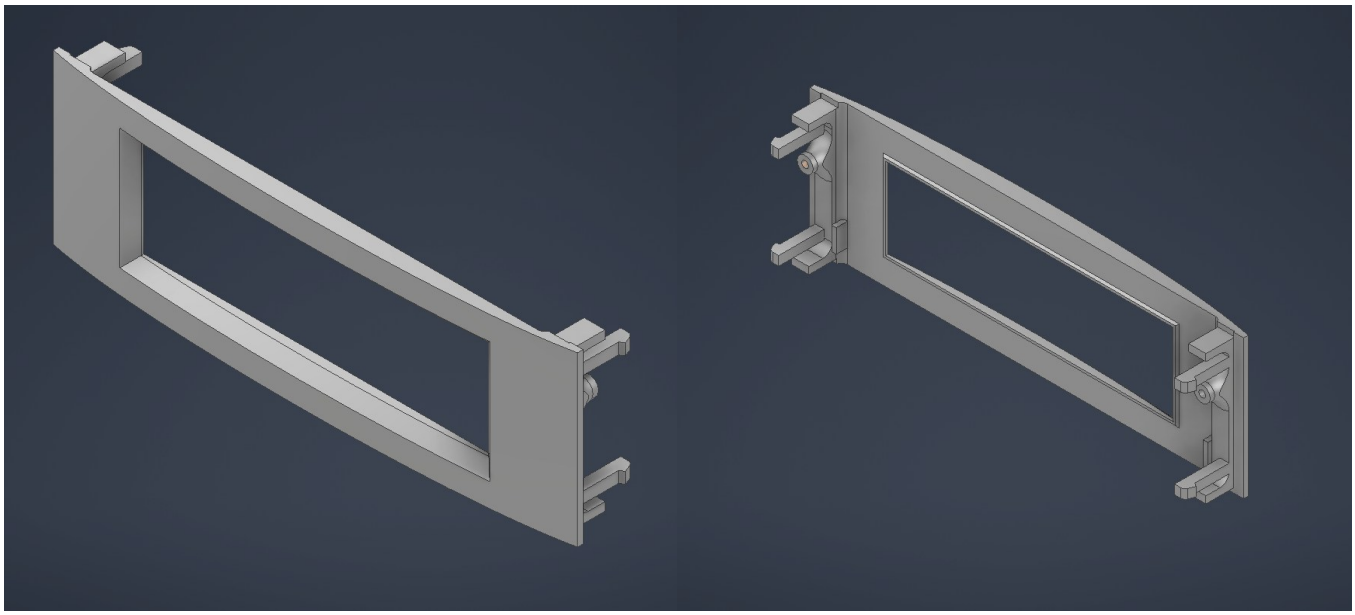


*Fig. 6. 3D CAD design of the custom dashboard adapter.*




*Fig. 7. Final physical integration and assembly of both screens inside the vehicle.*



*Fig. 8. Detailed view of the mechanical connection and mounting interface.*

These projects were physically manufactured, assembled, and iterated by me using my personal 3D printer and workshop. I frequently design and fabricate custom functional parts, enclosures, and mechanical repairs in my free time, and I would be glad to compile further examples of my hands-on prototyping work upon request.